Experiment 6: Champagne Tower

Aim: To determine how much champagne is present in a particular glass of a champagne tower after pouring a given amount of champagne into the top glass.

Algorithm

1. Start the program.
2. Read the amount of champagne poured into the top glass.
3. Read the required row and glass number.
4. Initialize all glasses to zero and pour the champagne into the top glass.
5. Check each glass row by row.
6. If a glass contains more than one cup, calculate the overflow.
7. Distribute half of the overflow to each of the two glasses directly below it.
8. Repeat until the required row is reached.
9. Display the champagne amount in the selected glass, limited to a maximum of one cup.
10. Stop the program.

C

```
#include <stdio.h>

int main() {
    double tower[101][101] = {0};
    double poured;
    int query_row, query_glass, i, j;
    printf("Enter amount of champagne poured: ");
    scanf("%lf", &poured);
    printf("Enter query row and glass: ");
    scanf("%d %d", &query_row, &query_glass);
    tower[0][0] = poured;
    for (i = 0; i < query_row; i++) {
        for (j = 0; j <= i; j++) {
            if (tower[i][j] > 1.0) {
                double overflow = (tower[i][j] - 1.0) / 2.0;
                tower[i + 1][j] += overflow;
                tower[i + 1][j + 1] += overflow;
            }
```

```
        }
    }

    if (tower[query_row][query_glass] > 1.0)
        printf("Champagne in glass: 1.000000\n");
    else
        printf("Champagne in glass: %.6f\n",
               tower[query_row][query_glass]);
    return 0;
}
```

Input:

Enter amount of champagne poured: 2

Enter query row: 1

Enter query glass: 1

Output:

```
Enter amount of champagne poured: 2
Enter query row and glass: 1
1
Champagne in glass: 0.500000
```

Result:

Thus, the C program successfully determines the amount of champagne present in a specified glass after pouring champagne into the top glass of the tower.